

Automatic Factor Grouping

Exposure deduplication and latent return factors

Abstract

This cluster-only run groups 140 admitted signals without known-factor labels. A training-only economic-admission stage retained 140 of 140 coverage-valid signals before HDBSCAN. HDBSCAN first collapses signals that rank the same stocks, including sign-reversed definitions. The resulting primitive FMP returns are then decomposed by parallel-analysis PCA and varimax rotation. The fresh version=20260829T051952Z sample produces 5 final candidate factors. Each parent gives every signed primitive composite one equal vote, regardless of how many raw signals that primitive contains. No post-clustering signal elimination or factor-admission rule is applied.

1 Sample and method

Setting	Value
Gold	version=20260829T051952Z
Requested symbols	all
Panel symbols	3014
Exposure window	2012-01-01 to 2022-12-31
Raw price floor	\$1.00
Market-cap screen	top 90.0% of names
Financials excluded	true
Minimum names / usable month	20
Minimum usable months	24
Pre-cluster Sharpe admission	one-shot point Sharpe
Full-period Sharpe cutoff	0.000
HDBSCAN minimum cluster size	2
Block-bootstrap repetitions	100
Primitive FMP sector demeaning	sector
Primitive FMP beta neutrality	false
Final factor FMP sector demeaning	sector
Final factor FMP beta neutrality	true
Parallel-analysis repetitions	200
Minimum absolute rotated loading	0.30
Minimum primitives / final factor	2
Parent construction	equal signed primitives

Monthly cross-sectional Spearman correlations are overlap-weighted through time. Exposure distance is $1 - |\rho|$, so a signal and its negation are treated as the same measurement and their sign is stored separately. HDBSCAN is used only for redundancy reduction. Final factors come from the common 131-month primitive-return window (2012-02-01 to 2022-12-01); no downloaded benchmark or known-factor return enters either stage. Beta is deliberately left in primitive returns so beta/BAB can appear as a latent risk dimension; beta neutrality belongs to later portfolio construction, not unsupervised grouping.

2 Training-only Sharpe admission

Each coverage-valid raw signal is first constructed as a standalone sector-demeaned, beta-neutral, signal-weighted long-short FMP after 10 bp turnover cost. Its direction is chosen from the 2012–2022 training Sharpe. It is admitted only when its single full-period oriented Sharpe is greater than 0.000. This makes the extension supervised;

no 2023+ return enters the screen.

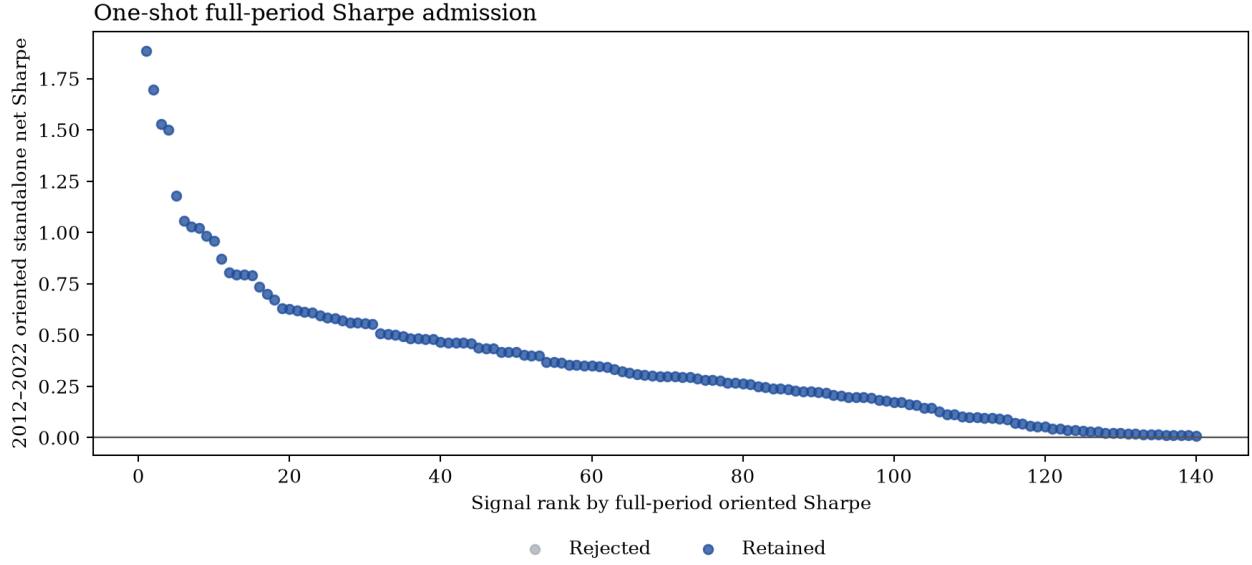


Figure 1: Ranked one-shot training Sharpe and the selected admission cutoff.

Rejected signal	N	Raw SR	Oriented SR	P(SR>0)	Reason
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3 Exposure-redundancy primitives

HDBSCAN produced 85 primitives: 41 multi-signal groups and 44 singletons. This is not treated as the final factor count; the earlier HDBSCAN-only interpretation failed because a dense redundancy map naturally preserves many unrelated signals as noise.

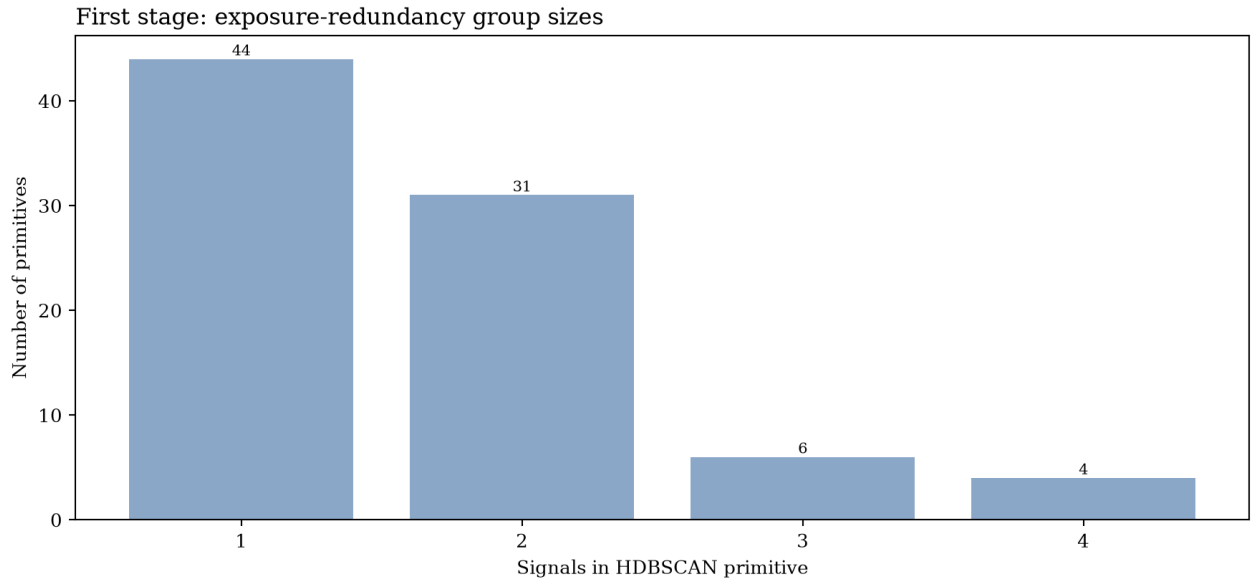


Figure 2: Distribution of signal counts in the exposure-redundancy primitives.

4 Latent factor count

Parallel analysis compares observed correlation-matrix eigenvalues with the 95th percentile from independently circular-shifted primitive returns. It retains 5 dimensions; 5 remain after the loading and minimum-signal rules.

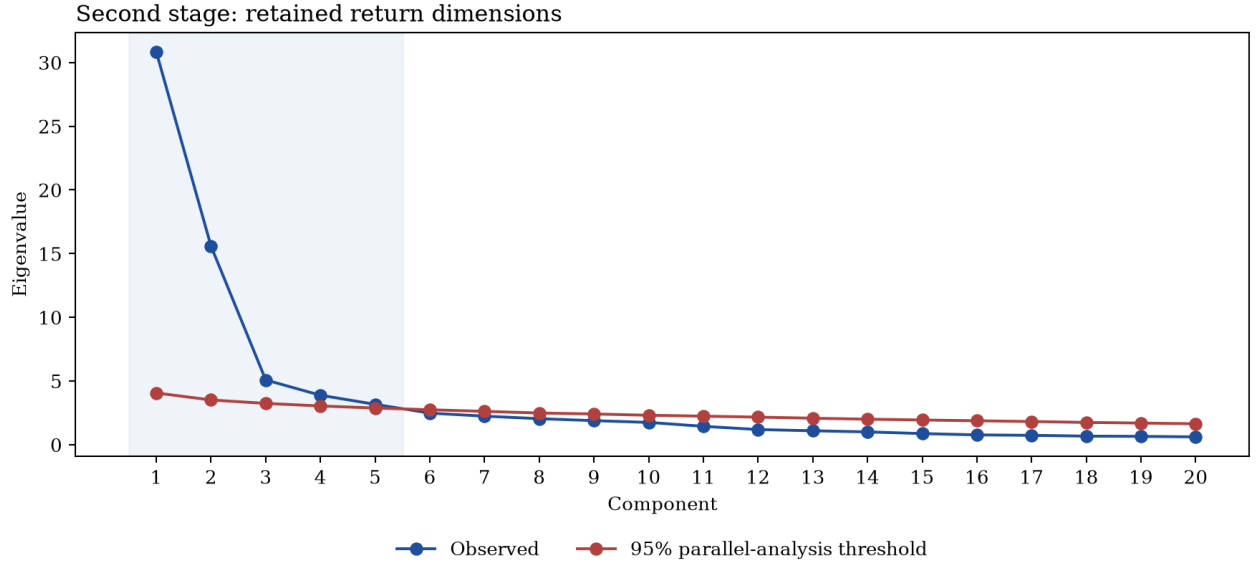


Figure 3: Observed and simulated eigenvalues. The shaded components are retained.

5 Rotated structure

Varimax rotation makes each primitive load mainly on one component. Loading sign orients a primitive within its factor; loading magnitude is not used as a portfolio weight. Each parent gives every signed primitive composite one equal vote, regardless of how many raw signals that primitive contains.

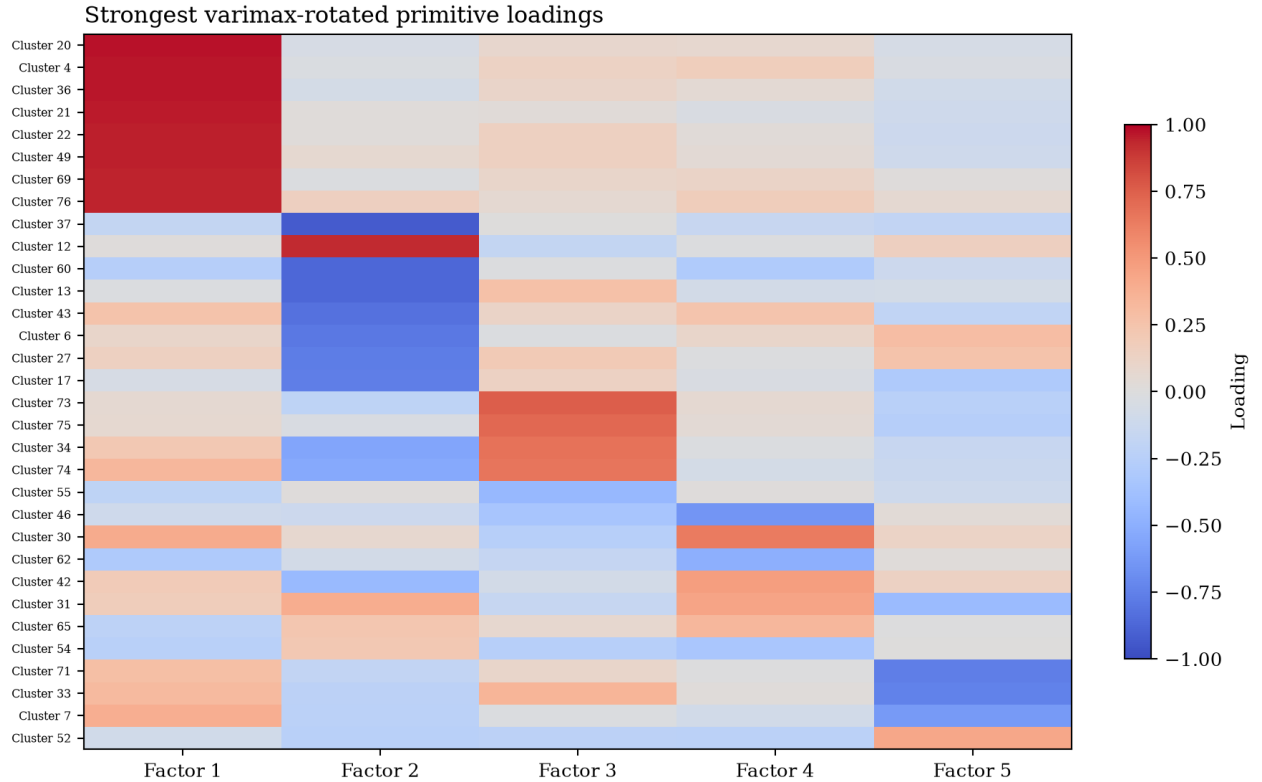


Figure 4: Rotated loading matrix, showing up to eight strongest retained primitives per factor.

6 Final candidate factors

The signs below give orientation only. When the Sharpe screen is enabled, economic evidence admits raw signals before clustering; no later rule removes a grouped signal or factor. The signed-signal table is the final equal-primitive construction.

Factor	Primitives	Raw signals	Signed primitive groups	Final signed signals
Factor 1	42	77	+Cluster 20, +Cluster 4, +Cluster 36, +Cluster 21, +Cluster 22, +Cluster 49, +Cluster 69, +Cluster 76, -Cluster 10, +Cluster 18, +Cluster 77, -Cluster 79, +Cluster 59, +Cluster 32, +Cluster 23, -Cluster 45, -Cluster 58, -Cluster 82, -Cluster 81, +Cluster 50, +Cluster 16, +Cluster 63, +Cluster 64, -Cluster 66, +Cluster 29, +Cluster 48, +Cluster 11, +Cluster 25, +Cluster 19, +Cluster 5, +Cluster 1, +Cluster 3, +Cluster 61, +Cluster 2, -Cluster 15, +Cluster 14, +Cluster 9, -Cluster 40, +Cluster 44, -Cluster 38, +Cluster 41, +Cluster 83	+bab 252v 1260c FRENCH CRSP, +bab shrunk 252v 1260c FRENCH CRSP, -beta 252v 1260c FRENCH CRSP, -beta shrunk 252v 1260c FRENCH CRSP, +bab 252v 1260c SPY, +bab shrunk 252v 1260c SPY, -beta 252v 1260c SPY, -beta shrunk 252v 1260c SPY, +bab 252v 1260c VTI, +bab shrunk 252v 1260c VTI, -beta 252v 1260c VTI, -beta shrunk 252v 1260c VTI, +operating profitabil- ity, +roa, +roe, +rota ebit, +accru- als, +accruals percent, +accruals to- tal, +eps growth yoy, +net income growth yoy, +net income growth yoy sign flip, -ivol 21d, -lottery max 5 21d, -vol 63d, -amihud illiq 21d, +dol- lar volume 21d, +buffett ronta proxy, +roc, -cash ratio, -cash to assets, +corr 1260d FRENCH CRSP, +corr 1260d VTI, +earnings yield, +ebit to ev, +ebit growth yoy ttm, +net income growth yoy ttm, +ebit margin ttm, +net income margin ttm, +fcf, +fcf margin, +fcf margin ttm, +fcf to eq- uity, +fcf to market cap, +fcf yield, +gp to equity, +gpoa, +growth roa 5y, +growth roe 5y, +log market cap, +market cap, +net margin, +operat- ing margin, -rd to opex, -rd to rev- enue, -ret volume corr 63d, -skew 63d, +surprise, +surprisePercentage, +as- set turnover, -beta stability 252d 252v 1260c SPY, +cash earnings ttm, +cfoa, +corr 1260d SPY, -equity issuance yoy, +ev, +fcf growth yoy ttm, +growth cfoa 5y, +high 52w 252d, -interest to debt, +is positive net income, +net in- come to market cap, +net operating as- sets proxy, -roe volatility, -sga to rev- enue, -share turnover 21d, +is negative equity
Factor 2	23	37	-Cluster 37, +Cluster 12, - Cluster 60, -Cluster 13, - Cluster 43, -Cluster 6, - Cluster 27, -Cluster 17, +Clus- ter 24, -Cluster 28, +Clus- ter 35, +Cluster 8, +Cluster 68, +Cluster 85, -Cluster 26, +Cluster 67, -Cluster 72, - Cluster 80, -Cluster 51, +Clus- ter 47, +Cluster 57, +Cluster 84, -Cluster 39	-accruals noa proxy, -asset growth yoy, -equity growth yoy, +debt to assets, +debt to equity, -ncav to mc, +assets to mc, -ev to assets, +book to mar- ket, -pb, +earnings to price, -pe ttm, +gp ev, +gross profit to ev, -gross mar- gin, -gross profit margin ttm, -gross profit growth yoy ttm, -revenue growth yoy, -growth gmar 5y, -growth gpoa 5y, -momentum 21 756, +value proxy 1260d, -ps ttm, +sales to price, -resid ret 252v 1260c SPY, -ret zscore 63d, - altman z, +capital intensity, -current ratio, +equity growth yoy sign flip, - evs ttm, +interest to opinc, +is nega- tive net operating assets, -momentum 126 252, -roic nopat, +is negative ev, +is negative invested capital

Factor	Primitives	Raw signals	Signed primitive groups	Final signed signals
Factor 3	5	6	+Cluster 73, +Cluster 75, +Cluster 34, +Cluster 74, -Cluster 55	+momentum 21 252, +resid momentum 21 252, -dso, +momentum 21 126, +momentum 21 252 vol 63d, +momentum 21 63
Factor 4	7	9	-Cluster 46, +Cluster 30, -Cluster 62, +Cluster 42, +Cluster 31, +Cluster 65, -Cluster 54	-is negative working capital, +working capital proxy, +is positive income tax, +tax rate, +accruals wc proxy, -capex to revenue, -dpo, -growth acc 5y, +info discreteness 21 252
Factor 5	4	7	-Cluster 71, -Cluster 33, -Cluster 7, +Cluster 52	-bb pctb 20d, -rsi 14d, -stoch k 14d, -ma dist 63d, -price zscore 63d, +debt issuance yoy, -momentum 0 21

7 Final factor FMP performance

Each factor is constructed once from its frozen component exposures using the configured rank transformation, composite-level sector demeaning, signal-weighted long-short portfolio and factor-level beta neutrality. The curves and statistics use the same complete common return window.

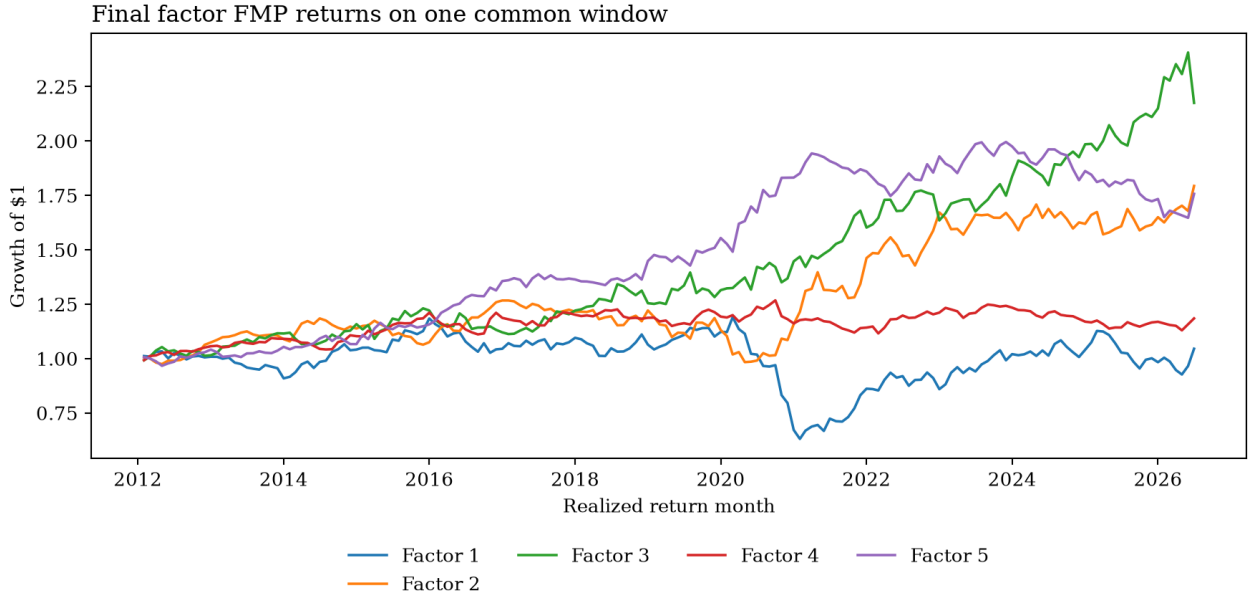


Figure 5: Net cumulative returns of the final factor books.

Factor	N	Start	End	Ann. return	Ann. vol.	Sharpe	HAC t	Max DD
Factor 1	174	2012-02-01	2026-07-01	0.98%	11.58%	0.09	0.29	-47.02%
Factor 2	174	2012-02-01	2026-07-01	4.44%	9.05%	0.49	1.69	-22.38%
Factor 3	174	2012-02-01	2026-07-01	5.77%	8.94%	0.64	3.31	-9.63%
Factor 4	174	2012-02-01	2026-07-01	1.29%	4.92%	0.26	1.06	-11.99%
Factor 5	174	2012-02-01	2026-07-01	4.12%	6.81%	0.60	2.42	-17.47%

8 Interpretation and reproducibility

This remains an exploratory return-risk taxonomy: the performance table describes the resulting books but does not select or eliminate factors. Exposure discovery run 20260906T015830Z_f4cd7c97ac. The run directory preserves primitive definitions, raw-signal lineage, rotated loadings, stock-level final exposures, primitive and final FMP returns, configuration and manifest for Factor Runner or HMM reuse.